

7(a)(i)	<i>area A</i> – intermembrane space ; <i>process B</i> – chemiosmosis / ATP synthesis ; <i>substance C</i> – oxygen ;	3
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Question	Answer	Marks
7(a)(ii)	enzyme / protein / polypeptide ;	1
7(a)(iii)	substrate-linked phosphorylation ;	1
7(b)(i)	22 ($\div$) 7 ; 3.14 ;	2
7(b)(ii)	any two from: 1 more red blood cells so more haemoglobin; 2 more haemoglobin to carry oxygen ; 3 more chance of aerobic respiration or - less chance of anaerobic respiration ;	2
7(c)	any two from: 1 (relatively) more C–H bonds ; 2 more reduced NAD produced ; 3 more hydrogen atoms to build up proton gradient ; 4 more, chemiosmosis / oxidative phosphorylation ;	2
7(d)(i)	(thermal) insulation ;	1

Question	Answer	Marks
7(d)(ii)	any two from: 1 (thermo)receptors in the skin detect change ; 2 <i>ref. to</i> threshold / all or nothing law ; 3 sensory neurone sends impulse (to hypothalamus) ;	2
7(d)(iii)	any two from: 1 increased, glycogenolysis / glucose available ; 2 (so) increased respiration ; 3 (so) more heat energy generated (and transferred to blood) ;	2